Assignment 1 – CSI 2132

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Exercise 1

Given:

- Store(storeid, sname, employee_number, city)
- Product(pid, pname, price)
- Supply(storeid, pid)

A

Result: storeid of stores with fewer than 100 employees or in city 'Ottawa'.

Relational algebra:

$$\pi_{\text{storeid}}\left(\sigma_{\text{employee_number}<100 \ \vee \ \text{city}='Ottawa'}(\text{Store})\right) \tag{1}$$

B

Result: sname of stores that supply the product with pname 'pencil'.

Relational algebra:

$$\pi_{\text{sname}}\left(\text{Store} \bowtie_{\text{Store.storeid}=\text{Supply.storeid}} \left(\text{Supply} \bowtie_{\text{Supply.pid}=\text{Product.pid}} \sigma_{\text{pname}='pencil'}(\text{Product})\right)\right) \tag{2}$$

C

Result: sname and city of stores that supply all products supplied by the store with storeid '0808'.

Let

$$R(\text{storeid}, \text{pid}) = \text{Supply} \tag{3}$$

$$P(\text{pid}) = \pi_{\text{pid}}(\sigma_{\text{storeid}='0808'}(\text{Supply})) \tag{4}$$

Expanded:

$$T = \pi_{\text{storeid}}(R) - \pi_{\text{storeid}}\left((\pi_{\text{storeid}}(R) \times P - R)\right) \tag{5}$$

Final result:

$$\pi_{\text{sname,city}}\left(\text{Store} \bowtie_{\text{Store.storeid}=T.\text{storeid}} T\right) \tag{6}$$

Exercise 2

Given:

- Employee(eid, cid, salary, managerid)
- Company(cid, companyname, location)
- Shares(eid, cid, sharenum)

A

Result: eid of employees who work for company 'Google' and have more than 500 shares in company 'Facebook'.

Relational algebra (short form with renaming):

$$\pi_{E.\text{eid}}\left(\sigma_{\substack{C_1.\text{companyname}='Google' \\ \wedge\, C_2.\text{companyname}='Facebook' \\ \wedge\, S.\text{sharenum}>500}}\left(E \bowtie_{E.\text{cid}=C_1.\text{cid}} \rho_{C_1}(C) \bowtie_{E.\text{eid}=S.\text{eid}} \left(S \bowtie_{S.\text{cid}=C_2.\text{cid}} \rho_{C_2}(C)\right)\right)\right) \tag{7}$$

B

Result: eid of employees whose manager owns shares in the company where the employee works.

Let

$$E_1 = \rho_{E_1}(\text{Employee}), \quad E_2 = \rho_{E_2}(\text{Employee}), \quad S = \text{Shares} \tag{8}$$

Relational algebra:

$$\pi_{E_1.\text{eid}}\left(\sigma_{E_1.\text{managerid}=E_2.\text{eid}\wedge E_2.\text{eid}=S.\text{eid}\wedge E_1.\text{cid}=S.\text{cid}}(E_1 \times E_2 \times S)\right) \tag{9}$$

C

Result: eid of employees who own shares in at least 3 different companies (no aggregation).

Let

$$S_1 = \rho_{S_1}(\text{Shares}), \quad S_2 = \rho_{S_2}(\text{Shares}), \quad S_3 = \rho_{S_3}(\text{Shares}) \tag{10}$$

Relational algebra:

$$\pi_{S_1.\text{eid}}\left(\sigma_{S_1.\text{eid}=S_2.\text{eid}\wedge S_1.\text{eid}=S_3.\text{eid}\wedge S_1.\text{cid}\neq S_2.\text{cid}\wedge S_1.\text{cid}\neq S_3.\text{cid}\wedge S_2.\text{cid}\neq S_3.\text{cid}}(S_1 \times S_2 \times S_3)\right) \tag{11}$$

D

Result: eid of employees who have shares in all companies in the database (no aggregation).

Let

$$R(\text{eid}, \text{cid}) = \pi_{\text{eid},\text{cid}}(\text{Shares}), \quad C(\text{cid}) = \pi_{\text{cid}}(\text{Company}) \tag{12}$$

Relational algebra using division:

$$\pi_{\text{eid}}(R \div C) \tag{13}$$

Expanded:

$$T = \pi_{\text{eid}}(R) - \pi_{\text{eid}}\left((\pi_{\text{eid}}(R) \times C - R)\right) \tag{14}$$

$$\pi_{\text{eid}}(T) \tag{15}$$

Exercise 3

Given:

- Supplier(supplier-id, name, city)
- Store(store-id, sname, city)
- Product(barcode, pname, price, itsshoes, itsabag, color, supplier-id)
- Has_stock(store-id, barcode, quantity)

A

Result: sname of stores that have at least two different black bag designs in stock (no aggregation).

Let

$$P_1 = \rho_{P_1}(\text{Product}), \quad P_2 = \rho_{P_2}(\text{Product}) \tag{16}$$

$$H_1 = \rho_{H_1}(\text{Has_stock}), \quad H_2 = \rho_{H_2}(\text{Has_stock}) \tag{17}$$

Relational algebra:

$$\pi_{\text{sname}}\left(\sigma_{\substack{P_1.\text{itsabag}='yes' \wedge P_1.\text{color}='black' \\ \wedge\, P_2.\text{itsabag}='yes' \wedge P_2.\text{color}='black' \\ \wedge\, H_1.\text{store-id}=H_2.\text{store-id} \\ \wedge\, H_1.\text{barcode}=P_1.\text{barcode} \\ \wedge\, H_2.\text{barcode}=P_2.\text{barcode} \\ \wedge\, H_1.\text{barcode}\neq H_2.\text{barcode}}} \left(\text{Store} \bowtie_{\text{Store.store-id}=H_1.\text{store-id}} (H_1 \times H_2 \times P_1 \times P_2)\right)\right) \tag{18}$$

B

Result: prices of all products that are shoes in the store named 'LaFollie'.

Relational algebra:

$$\pi_{\text{price}}\left(\sigma_{\text{sname}='LaFollie' \wedge \text{itsshoes}='yes'} \left(\text{Store} \bowtie_{\text{Store.store-id}=\text{Has_stock.store-id}} \text{Has_stock} \bowtie_{\text{Has_stock.barcode}=\text{Product.barcode}} \text{Product}\right)\right) \tag{19}$$

C

Result: name of suppliers that supply more than 5 products with different barcodes (aggregation allowed).

Let

$$R = \text{Supplier} \bowtie_{\text{Supplier.supplier-id}=\text{Product.supplier-id}} \text{Product} \tag{20}$$

$$G = \gamma_{\text{supplier-id,name,city; COUNT(barcode)}\rightarrow\text{prodcount}}(R) \tag{21}$$

Relational algebra:

$$\pi_{\text{name}}\left(\sigma_{\text{prodcount}>5}(G)\right) \tag{22}$$

D

Result: barcode of products that are sold both by store-id '1' and store-id '2' (no intersection operator).

Let

$$H_1 = \rho_{H_1}(\text{Has_stock}), \quad H_2 = \rho_{H_2}(\text{Has_stock}) \tag{23}$$

Relational algebra:

$$\pi_{H_1.\text{barcode}}\left(\sigma_{H_1.\text{store-id}='1' \wedge H_2.\text{store-id}='2' \wedge H_1.\text{barcode}=H_2.\text{barcode}}(H_1 \times H_2)\right) \tag{24}$$

Exercise 4

Given:

- Supplier(supplier-id, name, city)

- Store(store-id, sname, city)
- Product(barcode, pname, price, itsshoes, itsabag, color, supplier-id)
- Has_stock(store-id, barcode, quantity)

We use tuple relational calculus. Product(p) means tuple p is in relation Product, and similar for Store(s), Has_stock(h).

A

Result: all product tuples with price greater than 40.

Tuple relational calculus:

$$\{\, p \mid \mathrm{Product}(p) \wedge p.\,\mathrm{price} > 40 \,\} \tag{25}$$

B

Result: barcode of all products with price greater than 40.

Tuple relational calculus:

$$\{\, b \mid \exists p\, (\mathrm{Product}(p) \wedge p.\,\mathrm{price} > 40 \wedge b = p.\,\mathrm{barcode}) \,\} \tag{26}$$

C

Result: barcode of products sold in store-id '1' but not in store-id '2'.

Tuple relational calculus:

$$\begin{aligned} &\{\, b \mid \exists h_1\, (\mathrm{Has_stock}(h_1) \wedge h_1.\,\text{store-id} =' 1' \wedge b = h_1.\,\mathrm{barcode} \\ &\wedge \neg \exists h_2\, (\mathrm{Has_stock}(h_2) \wedge h_2.\,\text{store-id} =' 2' \wedge h_2.\,\mathrm{barcode} = h_1.\,\mathrm{barcode})) \,\} \end{aligned} \tag{27}$$

D

Result: barcode of products that are sold in all stores in city 'Ottawa'.

Tuple relational calculus:

$$\begin{aligned} &\{\, b \mid \exists p\big(\mathrm{Product}(p) \wedge b = p.\,\mathrm{barcode} \wedge \forall s\big(\mathrm{Store}(s) \wedge s.\,\mathrm{city} =' \mathit{Ottawa}' \\ &\rightarrow \exists h\big(\mathrm{Has_stock}(h) \wedge h.\,\text{store-id} = s.\,\text{store-id} \wedge h.\,\mathrm{barcode} = p.\,\mathrm{barcode}\big)\big)\big) \,\} \end{aligned} \tag{28}$$

References

- Standard database textbook (some)
- RelaX